KILNDECK · CONCEPT OPTIONS

KilnDeck concept options

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/02a-concept-options.md @ 9f92bd8		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-30	Amish Chadha	Option A counts, mass and cost updated for Amish's decisions of 2026-10-30; S23 now within known science
0.1	2026-10-28	Amish Chadha	First issue (kit 1.8 and 1.9 upgrade); option A is the current design; option B, a wall-top cableway, added; both modelled at D3; scored against KND-URS-001

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Decision: option A, the rolling bridge deck, which scores 152 of 190 against 124 for option B, the wall-top cableway (18 % apart). Amish decided on 2026-10-30, as recommended: "on the calls - i go with your recommendations" (KND-DEC-001 D10, KND-DDR-003, KND-DDR-004). Method: STANDARDS section 21.

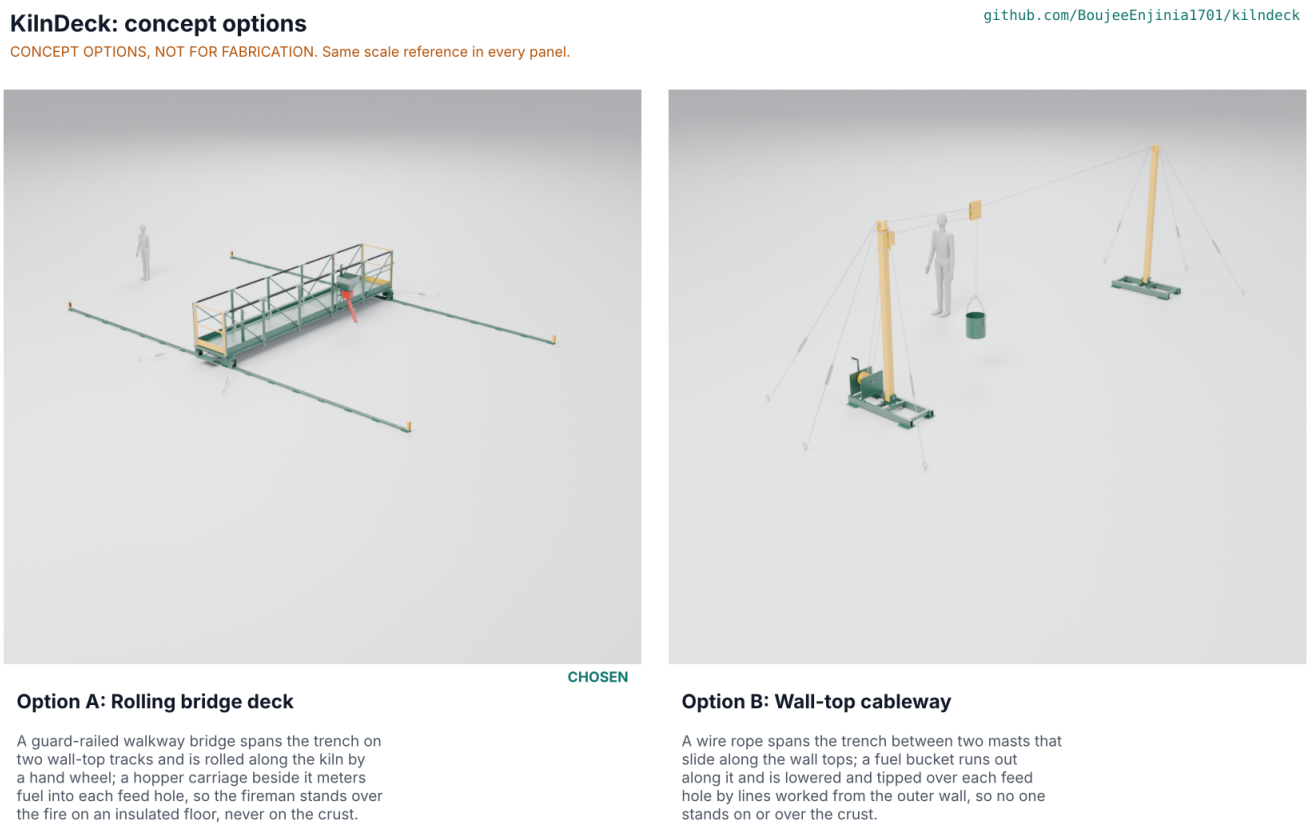


Figure 1. The two options, each drawn from its D3 model in `cad/src/model.py` at one scale with a 1.75 m person (rendered by `.kit/options.py` on Amish's Mac).

The options

Option A: rolling bridge deck (the current design)

A guard-railed walkway bridge spans the 6 m trench on two wall-top tracks and is rolled along the kiln by a hand wheel; a hopper carriage beside it meters fuel into each feed hole, so the fireman stands over the fire on an insulated floor, never on the crust. This is the design taken to TRL 3 on 2026-10-03 (KND-DDR-001, KND-DDR-002); the kit 1.9 upgrade models it part by part without changing its working principle. Key risks: floor heat in full sun (PR-007 not met), the wall bearing limit (survey), the 40 kg pieces (D14) and storms (fixed by D12). Needs from a co-design partner: a kiln survey, firemen's views on working over the fire, and space on the wall tops to stock fuel.

Option B: wall-top cableway

A wire rope spans the trench between two guyed masts that stand on skids on the wall tops. A trolley runs along the rope; a fuel bucket hangs from it on a lowering line, and the fireman, standing on the outer wall top beyond the skid, runs the trolley out with a haul line and lowers and tips the bucket over each feed hole with a two-drum hand winch. The skids slide 1 m along the walls for each row of holes. No one stands on or over the crust. Key risks: aiming a swinging bucket into a 150 mm hole by lines, dumping a whole bucket rather than a dose, resetting six guy anchors for every row, rope wear in heat and dust, and a heavy load hanging over the fire. Needs from a co-design partner: whether owners accept guy anchors in the yard beside the kiln, and a trial of aiming by line.

First-order numbers

Table 1. Options compared (all estimates; masses and part counts from `cad/fidelity.json`).

Quantity	Option A	Option B
Estimated parts cost (USD)	2,905 (BOM)	about 900 (estimate from the model mass and bought rope, sheaves, anchors and turnbuckles; not priced line by line)
Mass (kg)	1,265.5 (deck, carriage, 2 x 12 m of track, storm tie-downs)	260.0 (two masts and skids, guys, anchors, rope, trolley, bucket, winch)
Throughput	18 holes in a 17.5 min round, about 0.49 kg a pocket, 52 L hopper	about 52 s a hole against the 58 s the round allows (STM B4); 22 L bucket, about 10 holes a fill
Effort	68 N at the hand wheel rim; 12 N at the crank	about 470 N to slide a skid (two people); winch crank light
Parts count (made, bought, fasteners, welds)	2,111 parts (585 made or bought, 1,025 fasteners, 501 welds)	252 parts (103 made or bought, 113 fasteners, 36 welds)
STEM check	Open: 8 rows plausible (S3, S9, S11, S15, S18, S19, S20, S21); S17 and S22 fail with a fix proposed (KND-DEC-001 D16); S8 closed by a fix	Within known science on B1, B2, B5; B3 and B4 plausible; shares S1, S9 and S21

Scoring

Table 2. Weighted scoring (weights: High UR 3, Medium 2, Low 1; scores 1 to 5).

Criterion	Source	Weight	A	B	Reason
Off the crust	UR-001	3	5	5	Both keep the fireman off the crust
No fall	UR-002	3	5	3	A: 1.1 m guardrail all round; B: the fireman works at the trench edge of the wall top behind a short rail
Keep the round, fuel in the hole	UR-003	3	4	2	A: spout within 17 mm; B: aiming a swinging bucket by line, about 52 s a hole against 58 s
By hand	UR-004	3	5	4	Both manual; B needs two lines worked together
Less heat	UR-005	3	3	4	A stands over the fire (floor 74 °C in full sun); B on the wall top beside it
Fits my kiln	UR-007	3	4	3	A: pads at 0.154 MPa; B: mast skids plus six anchors in the yard per wall
Two-person carry	UR-008	3	3	3	A: 32.9 kg middle truss modules, 34.8 kg end modules (D14, D16); B: a 2.4 m SHS 100 x 5 mast is about 35 kg
Stays put	UR-013	3	4	3	A: lock pin, stops, storm clamps; B: the bucket hangs on a winch pawl over the fire
Steady dose	UR-006	2	4	2	A: metering rotor; B: the bucket is tipped by eye
Moves with the fire	UR-009	2	4	2	A: relay 2.4 m track modules; B: move both skids and reset six anchors every row
Affordable	UR-010	2	3	4	A USD 2,905; B about USD 900 (estimate)
Local build	UR-011	2	4	3	B needs wire rope terminations and a winch
Season-proof	UR-012	2	4	3	B: rope and guys wear in heat, dust and storms
Easy upkeep	UR-014	1	3	3	A: chains and many bolts; B: rope inspection
Safety overall	STANDARDS s21	3	4	4	A carries a person over the fire but guarded; B keeps people off the deck but hangs loads over it
Weighted total (of 190)			**152**	**124**	A 80 %, B 65 %

Cost against the value-engineering target, buildability and maintenance are scored through UR-010, UR-011 and UR-014.

Recommendation

Option A, decided by Amish on 2026-10-30 (KND-DEC-001 D10). It scores 18 % above option B, so it is not a choice within 10 %. Option A also carries the whole TRL 3 record (calculation note, drawings, build plan). The runner-up's best idea to graft onto it: a small wall-top cableway from the outer wall could serve the curved kiln ends, which option A does not reach (PR-001), or lift fuel to the hopper's refill point; both are suggestions for a later stage, not added here.